B.TECH. (COMPUTER ENGINEERING)

Semester 3:

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	BS	AMS2612	Higher Mathematics	3	1	0	4	15	25	60	100
2	PC	COC2032	Object Oriented Programming	3	1	0	4	15	25	60	100
3	PC	COC2062	Data Structure and Algorithm	3	1	0	4	15	25	60	100
4	PC	COC2072	Digital Logic and System Design	3	1	0	4	15	25	60	100
5	PC	COC2122	Database Management System	3	1	0	4	15	25	60	100
6	PC	COC2932	C & C++ Programming Lab	0	1	2	2	60		40	100
7	ESA	ELA2112	Electronic Devices & Circuits	3	1	0	4	15	25	60	100

TOTAL CREDITS: 26

Semester 4:

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	BS	AMS2622	Numerical Analysis, Transforms and Probability	3	0	0	3	15	25	60	100
2	BS	AMS2632	Discrete Structures	3	1	0	4	15	25	60	100
3	PC	COC2082	Computer Architecture	3	1	0	4	15	25	60	100
4	PC	COC2142	Design & Analysis of Algorithm	3	1	0	4	15	25	60	100
5	PC	COC2922	Digital Design & Simulation Lab	0	1	2	2	60	0	40	100
6	PC	COC2942	Data Structure Lab	0	1	2	2	60		40	100
7	ESA	ELA2412	Fundamentals of Digital Signal Processing	3	1	0	4	15	25	60	100
8	ESA	ELA2902	Electronics Laboratory	0	1	2	2	60	0	40	100

TOTAL CREDITS: 25

B.TECH. (COMPUTER ENGINEERING)

Semester 5:

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	PC	COC3092	Microprocessor Theory & Applications	3	0	0	3	15	25	60	100
2	PC	COC3102	Operating Systems	3	1	0	4	15	25	60	100
3	PC	COC3162	Software Engineering	3	0	0	3	15	25	60	100
4	PC	COC3932	Algorithms & Operating Systems Lab	0	1	2	2	60	0	40	100
5	PSI	COP3802	Colloquium	0	2	0	2	60		40	100
6	ESA	ELA3402	Communication Systems	3	1	0	4	15	25	60	100
7	HM	MEH3452	Engineering Economy & Management	3	0	0	3	15	25	60	100
8	OE		Open Elective-1				3	15	25	60	100

TOTAL CREDITS: 24

Semester 6:

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total
				L	T	P		Course-Work	Mid-Sem	End-Sem	
1	PC	COC3082	Digital Electronics	3	0	0	3	15	25	60	100
2	PC	COC3112	Theory of Computation	3	1	0	4	15	25	60	100
3	PC	COC3132	Computer Networks	3	1	0	4	15	25	60	100
4	PC	COC3152	Computer Graphics	3	0	0	3	15	25	60	100
5	PC	COC3962	Advanced Computing Lab	0	1	2	2	60	0	40	100
6	PC	COC3972	Microprocessor & Embedded Systems Lab	0	1	2	2	60		40	100
7	PSI	COP3952	Minor Project	0	0	3	3	60		40	100
8	HM		Elective Course under Humanities category				3	15	25	60	100
9	AU		Audit Course-2				0	15	25	60	100

TOTAL CREDITS: 24

DEPARTMENT OF COMPUTER ENGINEERING
Z.H. COLLEGE OF ENGINEERING & TECHNOLOGY
ALIGARH MUSLIM UNIVERSITY, ALIGARH

Course Structure: B.Tech. Computer Engineering

SEMESTER 7:

S.No	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total	Pre-Requisite Courses
				L	T	P		Course-Work	Mid-Sem	End-Sem		
1	DE	COE4xx0	Departmental Elective-I	3	1	0	4	15	25	60	100	
2	DE	COE4xx0	Departmental Elective-II	3	1	0	4	15	25	60	100	
3	DE	COE4xx0	Departmental Elective-III	3	1	0	4	15	25	60	100	
4	DC	COC4010	Information Security	3	1	0	4	15	25	60	100	AMS2630
5	DC	COC4060	Compiler Design	3	1	0	4	15	25	60	100	AMS2630 / COC3110
6	DC	COC4980	B.Tech Project Part-I	0	0	8	4	60		40	100	COC2090
7	DC	COC4950	Network & Embedded System Lab	0	1	2	2	60		40	100	
8												
9												
10												
						TOTAL CREDITS:	**26**					

SEMESTER 8:

S.No	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total	Pre-Requisite Courses
				L	T	P		Course-Work	Mid-Sem	End-Sem		
1	DE	COE4xx0	Departmental Elective-IV	3	1	0	4	15	25	60	100	
2	DE	COE4xx0	Departmental Elective-V	3	1	0	4	15	25	60	100	
3	DE	COE4xx0	Departmental Elective-VI	3	1	0	4	15	25	60	100	
4	OE	----	Open Elective - II	3	1	0	4	15	25	60	100	
5	DC	COC4990	B.Tech Project Part-II	0	0	12	6	60		40	100	COC2090
6												
7												
8												
9												
10												
						TOTAL CREDITS:	**22**					

Course Structure: B.Tech. **COMPUTER ENGINEERING**

DEPARTMENTAL ELECTIVE COURSES

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total	Pre-Requisite Courses
				L	T	P		Course-Work	Mid-Sem.	End-Sem.		
1	DE	COE4030	Design of Programming Languages	3	1	0	4	15	25	60	100	
2	DE	COE4300	Artificial Intelligence & Neural Network	3	1	0	4	15	25	60	100	
3	DE	COE4310	Internet Tools	3	1	0	4	15	25	60	100	
4	DE	COE4320	Image Processing	3	1	0	4	15	25	60	100	
5	DE	COE4360	Information Technology	3	1	0	4	15	25	60	100	
6	DE	COE4380	Selected Topics in Computer Application	3	1	0	4	15	25	60	100	
7	DE	COE4390	Advance Microprocessor System & Design	3	1	0	4	15	25	60	100	
8	DE	COE4410	Pattern Recognition & Computer Vision	3	1	0	4	15	25	60	100	
9	DE	COE4420	VLSI Design Tools & Techniques	3	1	0	4	15	25	60	100	
10	DE	COE4440	Soft Computing	3	1	0	4	15	25	60	100	
11	DE	COE4450	Network Security	3	1	0	4	15	25	60	100	
12	DE	COE4480	Embedded Systems	3	1	0	4	15	25	60	100	
13	DE	COE4490	Internet Protocol	3	1	0	4	15	25	60	100	
14	DE	COE4500	Multimedia Technologies	3	1	0	4	15	25	60	100	
15	DE	COE4510	Computer Network Design	3	1	0	4	15	25	60	100	
16	DE	COE4520	Mobile Computing	3	1	0	4	15	25	60	100	
17	DE	COE4530	Artificial Intelligence and Soft Computing	0	1	0	4	15	25	60	100	
18	DE	COE4540	Software Testing	3	1	0	4	15	25	60	100	
19	DE	COE4550	Real Time Systems	3	1	0	4	15	25	60	100	
20	DE	COE4560	Big Data Analytics	3	1	0	4	15	25	60	100	
21	DE	COE4570	Internet of Things	3	1	0	4	15	25	60	100	
22	DE	COE4580	Cloud Computing	3	1	0	4	15	25	60	100	
23	DE	COE4590	Distributed & Parallel Systems	3	1	0	4	15	25	60	100	
24	DE	COE4600	Principles of Cyber Physical Systems	3	1	0	4	15	25	60	100	
25	DE	COE4610	Principles of Machine Learning	3	1	0	4	15	25	60	100	
26	DE	COE4620	Bioinformatics	3	1	0	4	15	25	60	100	
27	DE	COE4630	Human Centric Computing	3	1	0	4	15	25	60	100	
28	DE	COE4640	Scientometrics	3	1	0	4	15	25	60	100	

OPEN ELECTIVE COURSES

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total	Pre-Requisite Courses
				L	T	P		Course-Work	Mid-Sem.	End-Sem.		
1	OE	COO4460	Selected Topics in Computer Engineering-I	3	1	0	4	15	25	60	100	
2	OE	COO4470	Selected Topics in Computer Engineering-II	3	1	0	4	15	25	60	100	
2	OE	COO4330	Smart Grids	3	1	0	4	15	25	60	100	

COURSE(S) FOR ELECTRONICS ENGINEERING DEPARTMENT

S.No.	Course Category	Course No.	Course title	Contact Periods			Credits	Marks			Total	Pre-Requisite Courses
				L	T	P		Course-Work	Mid-Sem.	End-Sem.		
1	ESA	COA3600	Data Structures & Programming	3	0	0	3	15	25	60	100	